

SR

SRFashionStyle

Perancangan Basis Data

Isi: ERD · Transformasi ERD · LRS · Kamus Data · Relasi Tabel | Tanggal: 23 Juni 2026

15

TABEL

6

MODUL BISNIS

25

FILE MIGRASI

Perancangan Basis Data — SRFashionStyle

- > **Tujuan:** Dokumentasi perancangan basis data untuk laporan akhir Kerja Praktek
 - > **Cakupan:** Seluruh tabel pada sistem backend Laravel
 - > **Tools:** MySQL / MariaDB, Laravel Migration
-

1. Entity Relationship Diagram (ERD)

erDiagram

users ||--o{ orders : "memiliki (nullable)"

users ||--o{ reviews : "menulis"

users ||--o{ email_otp : "meminta (nullable)"

users ||--o{ pos_shifts : "membuka"

users ||--o{ personal_access_tokens : "token"

categories ||--o{ products : "berisi"

products ||--o{ product_variants : "memiliki"

products ||--o{ product_images : "memiliki"

products ||--o{ order_items : "terdapat di"

products ||--o{ reviews : "diulas"

product_variants ||--o{ inventories : "stok"

product_variants ||--o{ order_items : "dibeli via"

product_variants ||--o{ pos_transaction_items : "dijual via"

orders ||--o{ order_items : "terdiri dari"

orders ||--o{ reviews : "direview"

cashier_staff ||--o{ pos_shifts : "bertugas"

cashier_staff ||--o{ pos_transactions : "melayani"

pos_shifts ||--o{ pos_transactions : "mencatat"

pos_transactions ||--o{ pos_transaction_items : "terdiri dari"

users {

bigint id PK

```
    string name

    string email UK

    enum role "owner,admin,cashier,customer"

    string phone

    text address

    string city

    string postal_code

    timestamp email_verified_at

    string password

    string remember_token

    timestamp created_at

    timestamp updated_at
}

categories {

    bigint id PK

    string name

    string slug UK

    boolean is_active

    timestamp created_at

    timestamp updated_at
}

products {

    bigint id PK

    bigint category_id FK
```

```
    string name

    string slug UK

    text description

    decimal base_price

    string featured_image

    boolean is_active

    timestamp created_at

    timestamp updated_at
}

product_variants {

    bigint id PK

    bigint product_id FK

    string sku UK

    string color

    string size

    decimal price

    boolean is_active

    timestamp created_at

    timestamp updated_at
}

product_images {

    bigint id PK

    bigint product_id FK

    string file_path
```

```
    string file_name

    int sort_order

    boolean is_primary

    timestamp created_at

    timestamp updated_at
}

inventories {

    bigint id PK

    bigint product_variant_id FK

    int stock_on_hand

    int stock_reserved

    int stock_available

    int min_stock_alert

    timestamp created_at

    timestamp updated_at
}

orders {

    bigint id PK

    string order_code UK

    string customer_name

    string email

    string phone

    text address

    string city
```

```
string postal_code

string shipping_method

string shipping_service_code

string shipping_label

string payment_method

text notes

enum channel "online,pos"

string pos_payment_method

string pos_transfer_ref

decimal subtotal

decimal shipping_cost

decimal grand_total

string status

string payment_status

string payment_gateway

string payment_reference

string biteship_order_id

string tracking_id

string waybill_id

string shipping_status

timestamp paid_at

bigint user_id FK "nullable"

timestamp created_at

timestamp updated_at

}
```

```
order_items {  
  
    bigint id PK  
  
    bigint order_id FK  
  
    bigint product_id FK  
  
    bigint product_variant_id FK  
  
    string product_name  
  
    string sku  
  
    string color  
  
    string size  
  
    decimal price  
  
    int qty  
  
    decimal subtotal  
  
    timestamp created_at  
  
    timestamp updated_at  
  
}
```

```
reviews {  
  
    bigint id PK  
  
    bigint order_id FK  
  
    bigint order_item_id FK  
  
    bigint product_id FK  
  
    bigint user_id FK  
  
    int rating  
  
    text comment  
  
    timestamp created_at
```



```
        timestamp updated_at
    }

    cashier_staff {

        bigint id PK

        string name

        string code UK

        boolean is_active

        text notes

        timestamp created_at

        timestamp updated_at
    }

    pos_shifts {

        bigint id PK

        bigint cashier_staff_id FK

        bigint opened_by_user_id FK

        string shift_code UK

        decimal opening_cash

        decimal closing_cash

        decimal total_cash_in

        decimal total_non_cash

        decimal cash_difference

        enum status "open,closed"

        timestamp opened_at

        timestamp closed_at
    }
}
```

```
    text notes

    timestamp created_at

    timestamp updated_at
}

pos_transactions {

    bigint id PK

    bigint pos_shift_id FK

    bigint cashier_staff_id FK

    string transaction_code UK

    decimal subtotal

    decimal discount

    decimal grand_total

    enum payment_method "cash,qris,transfer_bca,card"

    decimal cash_received

    decimal change_given

    string transfer_ref

    enum status "completed,cancelled,refunded"

    text notes

    timestamp created_at

    timestamp updated_at
}

pos_transaction_items {

    bigint id PK

    bigint pos_transaction_id FK
```

```
    bigint product_variant_id FK

    string product_name

    string sku

    string color

    string size

    decimal price

    int qty

    decimal subtotal

    timestamp created_at

    timestamp updated_at
}

email_otp {

    bigint id PK

    bigint user_id FK "nullable"

    string email

    string purpose

    string code_hash

    int attempt_count

    timestamp expires_at

    timestamp last_sent_at

    timestamp verified_at

    timestamp created_at

    timestamp updated_at
}
```

```
promotion_media {  
  
    bigint id PK  
  
    string title  
  
    string placement  
  
    string active_for  
  
    string image_path  
  
    text image_url  
  
    timestamp created_at  
  
    timestamp updated_at  
  
}  
  
personal_access_tokens {  
  
    bigint id PK  
  
    string tokenable_type  
  
    bigint tokenable_id  
  
    string name  
  
    string token UK  
  
    text abilities  
  
    timestamp last_used_at  
  
    timestamp expires_at  
  
    timestamp created_at  
  
    timestamp updated_at  
  
}  
  
password_reset_tokens {  
  
    string email PK
```

```
        string token

        timestamp created_at
    }

    failed_jobs {

        bigint id PK

        string uuid UK

        text connection

        text queue

        longText payload

        longText exception

        timestamp failed_at
    }
```

2. Transformasi ERD ke Skema Relasional

Proses transformasi dari model konseptual (ERD) ke skema relasional mengikuti aturan berikut:

Aturan 1: Setiap Entitas Kuat → Satu Tabel

Setiap entitas pada ERD ditransformasikan menjadi satu tabel dengan primary key dari atribut kunci entitas tersebut.

ENTITAS	TABEL	PRIMARY KEY
User	users	id
Category	categories	id
Product	products	id
ProductVariant	product_variants	id
ProductImage	product_images	id
Inventory	inventories	id
Order	orders	id
OrderItem	order_items	id
Review	reviews	id
CashierStaff	cashier_staff	id

ENTITAS	TABEL	PRIMARY KEY
PosShift	pos_shifts	id
PosTransaction	pos_transactions	id
PosTransactionItem	pos_transaction_items	id
EmailOtp	email_otps	id
PromotionMedia	promotion_media	id

Aturan 2: Relasi 1:N — Tambahkan FK di Sisi "Banyak"

Untuk setiap relasi *one-to-many*, foreign key ditambahkan pada tabel yang bersisi "banyak" (N), merujuk ke primary key tabel sisi "satu" (1).

RELASI	FK PADA TABEL	MERUJUK KE
User 1—N Order	orders.user_id (nullable)	users.id
User 1—N Review	reviews.user_id	users.id
User 1—N PosShift	pos_shifts.opened_by_user_id	users.id
Category 1—N Product	products.category_id	categories.id
Product 1—N ProductVariant	product_variants.product_id	products.id
Product 1—N ProductImage	product_images.product_id	products.id
ProductVariant 1—N Inventory	inventories.product_variant_id	product_variants.id
Order 1—N OrderItem	order_items.order_id	orders.id
CashierStaff 1—N PosShift	pos_shifts.cashier_staff_id	cashier_staff.id
CashierStaff 1—N PosTransaction	pos_transactions.cashier_staff_id	cashier_staff.id
PosShift 1—N PosTransaction	pos_transactions.pos_shift_id	pos_shifts.id
PosTransaction 1—N PosTransactionItem	pos_transaction_items.pos_transaction_id	pos_transactions.id

Aturan 3: Relasi M:N — Tabel Asosiasi

Relasi many-to-many antara `orders` dan `product_variants` dipecah melalui tabel asosiasi `order_items`. Tabel asosiasi ini memiliki dua FK yang masing-masing merujuk ke tabel induk:

TABEL ASOSIASI	FK 1	FK 2	FK 3 (KONTEKS)
order_items	order_id → orders.id	product_variant_id → product_variants.id	product_id → products.id
pos_transaction_items	pos_transaction_id → pos_transactions.id	product_variant_id → product_variants.id	—
reviews	order_id → orders.id	user_id → users.id	order_item_id → order_items.id

Aturan 4: Relasi 1:1 — FK Unik atau Merge

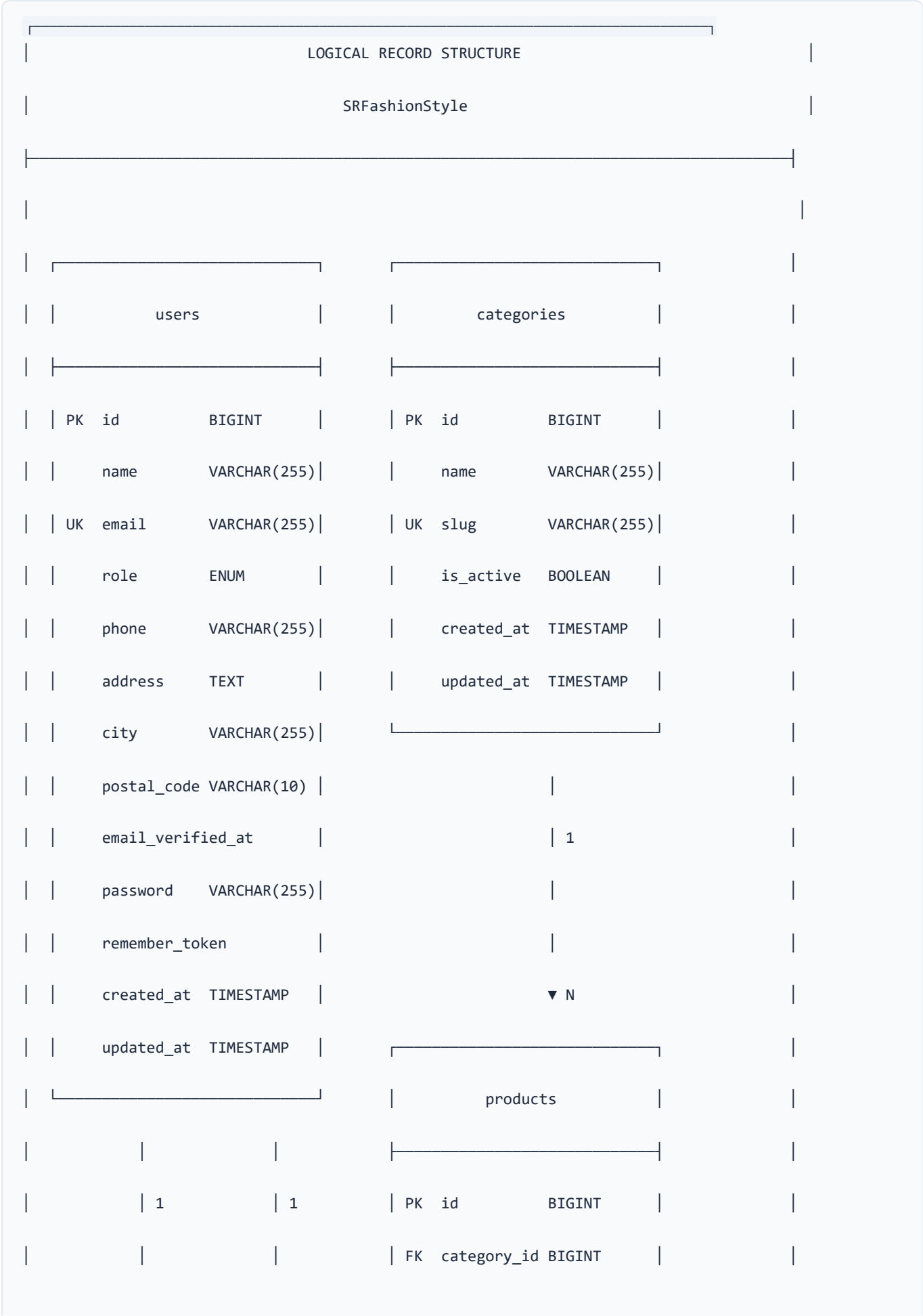
Pada sistem ini tidak terdapat relasi *one-to-one* murni. Relasi `orders` ke `pos_transactions` bersifat opsional (satu order online mungkin tidak memiliki transaksi POS, dan satu transaksi POS selalu merujuk ke satu order *pay-at-store*). Implementasi dilakukan melalui query aplikasi, bukan constraint database.

Aturan 5: Atribut Multivalue → Tabel Terpisah

Atribut `images` pada entitas `Product` yang bersifat multivalue ditransformasikan menjadi tabel `product_images` dengan FK ke `products`.

3. Logical Record Structure (LRS)

LRS menyajikan struktur tabel akhir setelah transformasi, dengan hanya menampilkan: nama tabel, nama kolom, tipe data, primary key (PK), foreign key (FK), dan unique key (UK).

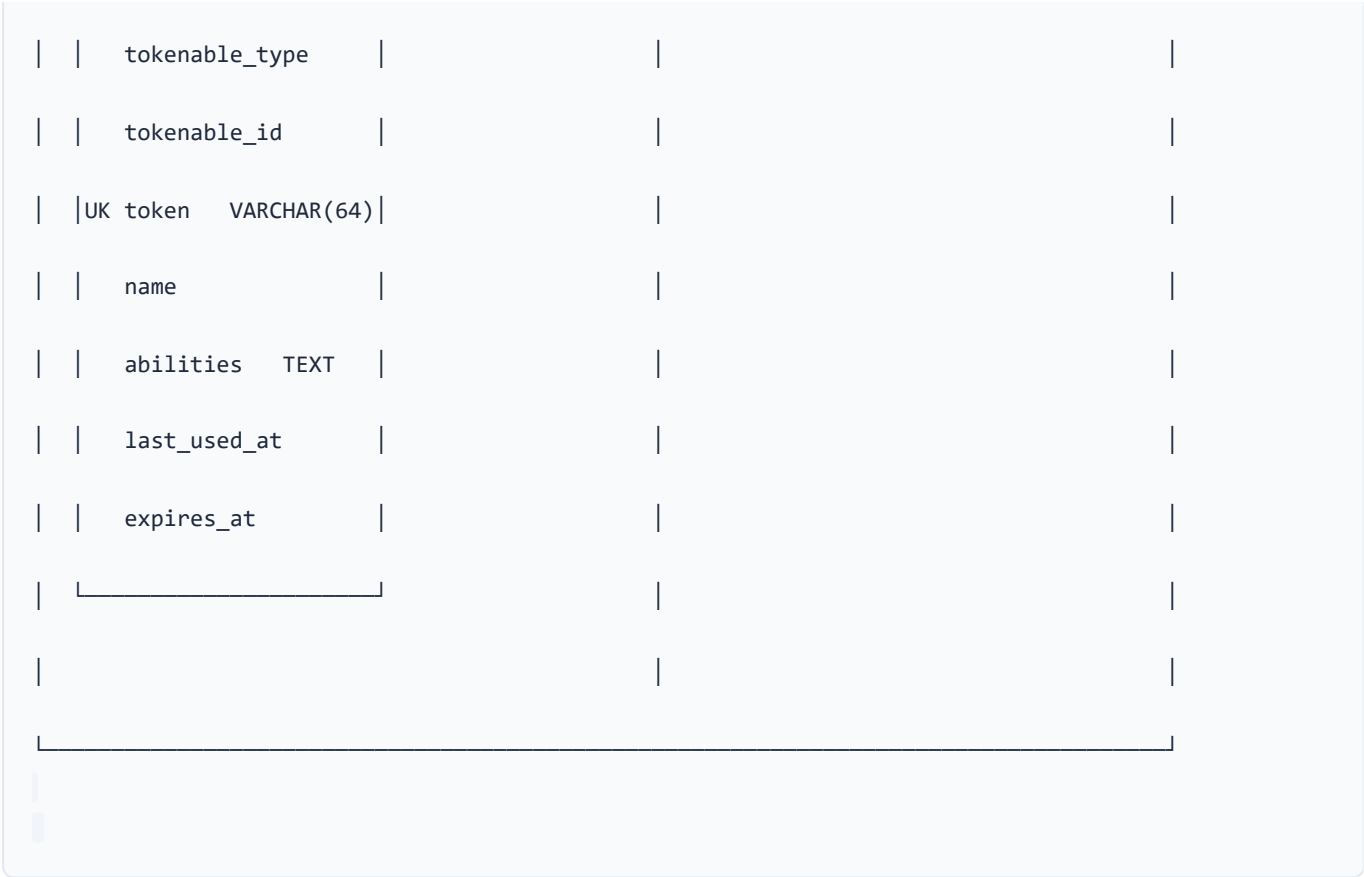


	▼ N	▼ N	name	VARCHAR(255)		
			UK slug	VARCHAR(255)		
orders	email_otps		description	TEXT		
			base_price	DECIMAL(12,2)		
PK id	BIGINT	PK id	BIGINT	featured_image		
UK order_code		FK user_id		is_active	BOOLEAN	
customer_name	email		created_at	TIMESTAMP		
email	purpose		updated_at	TIMESTAMP		
phone	code_hash					
address	TEXT	attempt_cnt				
city		expires_at	1	1	1	
postal_code		last_sent				
shipping_method	verified_at	▼ N	▼ N	▼ N		
payment_method	created_at					
channel	ENUM	updated_at	product_	product	order_items	
subtotal			images	variants		
shipping_cost					PK id	BIGINT
grand_total			PK id	PK id	FK order_id	
status			FK prod_id	FK prod	FK product_id	
payment_status			file_path	_id	FK pr_var_id	
payment_gateway			file_name	UK sku	product_name	
payment_ref			sort_order	color	sku	
biteship_ord_id			is_primary	size	color	
tracking_id			created_at	price	size	
waybill_id			updated_at	is_act	price	

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notes	TEXT	PK id	BIGINT
created_at		title	
updated_at		placement	
		active_for	
		image_path	
	1	image_url	
		created_at	
	▼ N	updated_at	
pos_transaction_items			
PK id	BIGINT		
FK pos_tr_id			
FK product_variant_id			
product_name			
sku			
color			
size			
price	DECIMAL		
qty	INT		
subtotal	DECIMAL		
created_at			
updated_at			





4. Kamus Data (Data Dictionary)

4.1 Tabel `users`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik user
name	VARCHAR(255)	NOT NULL	Nama lengkap
email	VARCHAR(255)	NOT NULL, UNIQUE	Alamat email (lowercase)
role	ENUM('owner','admin','cashier','customer')	NOT NULL, DEFAULT 'customer'	Peran pengguna
phone	VARCHAR(255)	NULLABLE	Nomor telepon
address	TEXT	NULLABLE	Alamat rumah
city	VARCHAR(255)	NULLABLE	Kota
postal_code	VARCHAR(10)	NULLABLE	Kode pos
email_verified_at	TIMESTAMP	NULLABLE	Waktu verifikasi email
password	VARCHAR(255)	NOT NULL	Hash password (bcrypt)
remember_token	VARCHAR(100)	NULLABLE	Token "remember me"
created_at	TIMESTAMP	NULLABLE	Waktu pendaftaran
updated_at	TIMESTAMP	NULLABLE	Waktu pembaruan terakhir

4.2 Tabel `categories`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik kategori
name	VARCHAR(255)	NOT NULL	Nama kategori
slug	VARCHAR(255)	NOT NULL, UNIQUE	URL-friendly slug
is_active	BOOLEAN	NOT NULL, DEFAULT true	Status aktif
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

4.3 Tabel `products`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik produk
category_id	BIGINT UNSIGNED	FK → categories(id), NOT NULL	Kategori produk
name	VARCHAR(255)	NOT NULL	Nama produk
slug	VARCHAR(255)	NOT NULL, UNIQUE	URL-friendly slug
description	TEXT	NULLABLE	Deskripsi produk
base_price	DECIMAL(12,2)	NOT NULL, DEFAULT 0	Harga dasar
featured_image	VARCHAR(255)	NULLABLE	Gambar utama
is_active	BOOLEAN	NOT NULL, DEFAULT true	Status aktif
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: `category_id` → ON UPDATE CASCADE, ON DELETE RESTRICT

4.4 Tabel `product_variants`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik varian
product_id	BIGINT UNSIGNED	FK → products(id), NOT NULL	Produk induk
sku	VARCHAR(255)	NOT NULL, UNIQUE	Kode SKU
color	VARCHAR(50)	NULLABLE	Warna
size	VARCHAR(20)	NULLABLE	Ukuran
price	DECIMAL(12,2)	NOT NULL	Harga varian
is_active	BOOLEAN	NOT NULL, DEFAULT true	Status aktif
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: `product_id` → ON DELETE CASCADE

4.5 Tabel `product_images`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik gambar

KOLOM	TIPE	CONSTRAINT	KETERANGAN
product_id	BIGINT UNSIGNED	FK → products(id), NOT NULL	Produk terkait
file_path	VARCHAR(255)	NOT NULL	Path file di storage
file_name	VARCHAR(255)	NULLABLE	Nama file asli
sort_order	INT	NOT NULL, DEFAULT 0	Urutan tampil
is_primary	BOOLEAN	NOT NULL, DEFAULT false	Gambar utama?
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: product_id → ON DELETE CASCADE

4.6 Tabel inventories

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik inventory
product_variant_id	BIGINT UNSIGNED	FK → product_variants(id), NOT NULL	Varian terkait
stock_on_hand	INT	NOT NULL, DEFAULT 0	Stok fisik
stock_reserved	INT	NOT NULL, DEFAULT 0	Stok dipesan (belum dibayar)
stock_available	INT	NOT NULL, DEFAULT 0	Stok tersedia → stock_on_hand - stock_reserved
min_stock_alert	INT	NOT NULL, DEFAULT 0	Batas minimum stok
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: product_variant_id → ON DELETE CASCADE

4.7 Tabel orders

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik order
order_code	VARCHAR(255)	NOT NULL, UNIQUE	Kode order (generated)
customer_name	VARCHAR(255)	NOT NULL	Nama pemesan
email	VARCHAR(255)	NOT NULL	Email pemesan
phone	VARCHAR(255)	NOT NULL	Telepon pemesan
address	TEXT	NOT NULL	Alamat pengiriman
city	VARCHAR(255)	NOT NULL	Kota pengiriman
postal_code	VARCHAR(255)	NOT NULL	Kode pos pengiriman
shipping_method	VARCHAR(255)	NULLABLE	jnt / pickup
shipping_service_code	VARCHAR(255)	NULLABLE	Kode layanan J&T
shipping_label	VARCHAR(255)	NULLABLE	Label layanan
payment_method	VARCHAR(255)	NOT NULL	Metode pembayaran
notes	TEXT	NULLABLE	Catatan
channel	ENUM('online','pos')	NOT NULL, DEFAULT 'online'	Channel penjualan
pos_payment_method	VARCHAR(50)	NULLABLE	Metode bayar POS (cash/qrisk/transfer_bca)
pos_transfer_ref	VARCHAR(100)	NULLABLE	Nomor referensi transfer (POS)

KOLOM	TIPE	CONSTRAINT	KETERANGAN
subtotal	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Subtotal item
shipping_cost	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Ongkos kirim
grand_total	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Total akhir
status	VARCHAR(255)	NOT NULL, DEFAULT 'pending'	pending / processing / shipped / completed / cancelled
payment_status	VARCHAR(255)	NOT NULL, DEFAULT 'pending'	pending / paid / failed / expired
payment_gateway	VARCHAR(255)	NULLABLE	midtrans
payment_reference	VARCHAR(255)	NULLABLE	Referensi dari gateway
biteship_order_id	VARCHAR(255)	NULLABLE	ID order di Biteship
tracking_id	VARCHAR(255)	NULLABLE	ID tracking Biteship
waybill_id	VARCHAR(255)	NULLABLE	Nomor resi
shipping_status	VARCHAR(255)	NULLABLE	Status dari Biteship
paid_at	TIMESTAMP	NULLABLE	Waktu pembayaran sukses
user_id	BIGINT UNSIGNED	FK → users(id), NULLABLE, ON DELETE SET NULL	User terautentikasi
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

4.8 Tabel `order_items`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik item
order_id	BIGINT UNSIGNED	FK → orders(id), NOT NULL	Order terkait
product_id	BIGINT UNSIGNED	FK → products(id), NOT NULL	Produk
product_variant_id	BIGINT UNSIGNED	FK → product_variants(id), NOT NULL	Varian yang dipilih
product_name	VARCHAR(255)	NOT NULL	Snapshot nama produk
sku	VARCHAR(255)	NOT NULL	Snapshot SKU
color	VARCHAR(255)	NULLABLE	Snapshot warna
size	VARCHAR(255)	NULLABLE	Snapshot ukuran
price	DECIMAL(15,2)	NOT NULL	Harga satuan saat order
qty	INT	NOT NULL	Jumlah
subtotal	DECIMAL(15,2)	NOT NULL	price × qty
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: `order_id`, `product_id`, `product_variant_id` → ON DELETE CASCADE

4.9 Tabel `reviews`

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik ulasan
order_id	BIGINT UNSIGNED	FK → orders(id), NOT NULL	Order terkait
order_item_id	BIGINT UNSIGNED	FK → order_items(id), NOT NULL	Item yang diulas

KOLOM	TIPE	CONSTRAINT	KETERANGAN
product_id	BIGINT UNSIGNED	FK → products(id), NOT NULL	Produk yang diulas
user_id	BIGINT UNSIGNED	FK → users(id), NOT NULL	Pemberi ulasan
rating	TINYINT UNSIGNED	NOT NULL	Rating 1–5
comment	TEXT	NULLABLE	Isi ulasan
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

Unique: (order_item_id , user_id) — satu ulasan per item per user **FK Cascade:** Semua FK → ON DELETE CASCADE

4.10 Tabel cashier_staff

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik kasir
name	VARCHAR(100)	NOT NULL	Nama kasir
code	VARCHAR(20)	NOT NULL, UNIQUE	Kode unik kasir
is_active	BOOLEAN	NOT NULL, DEFAULT true	Status aktif
notes	TEXT	NULLABLE	Catatan
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

4.11 Tabel pos_shifts

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik shift
cashier_staff_id	BIGINT UNSIGNED	FK → cashier_staff(id), NOT NULL	Kasir bertugas
opened_by_user_id	BIGINT UNSIGNED	FK → users(id), NOT NULL	User pembuka shift
shift_code	VARCHAR(50)	NOT NULL, UNIQUE	Kode shift (generated)
opening_cash	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Saldo awal
closing_cash	DECIMAL(15,2)	NULLABLE	Saldo akhir (diisi saat close)
total_cash_in	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Total transaksi tunai
total_non_cash	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Total transaksi non-tunai
cash_difference	DECIMAL(15,2)	NULLABLE	Selisih → closing_cash - (opening_cash + total_cash_in)
status	ENUM('open','closed')	NOT NULL, DEFAULT 'open'	Status shift
opened_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Waktu buka
closed_at	TIMESTAMP	NULLABLE	Waktu tutup
notes	TEXT	NULLABLE	Catatan
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

4.12 Tabel pos_transactions

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik transaksi
pos_shift_id	BIGINT UNSIGNED	FK → pos_shifts(id), NOT NULL	Shift terkait
cashier_staff_id	BIGINT UNSIGNED	FK → cashier_staff(id), NOT NULL	Kasir pelaksana
transaction_code	VARCHAR(50)	NOT NULL, UNIQUE	Kode transaksi (generated)
subtotal	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Subtotal item
discount	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Diskon
grand_total	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Total akhir
payment_method	ENUM('cash','qris','transfer_bca','card')	NOT NULL, DEFAULT 'cash'	Metode bayar
cash_received	DECIMAL(15,2)	NULLABLE	Uang diterima (cash)
change_given	DECIMAL(15,2)	NULLABLE	Kembalian (cash)
transfer_ref	VARCHAR(100)	NULLABLE	Nomor referensi transfer
status	ENUM('completed','cancelled','refunded')	NOT NULL, DEFAULT 'completed'	Status transaksi
notes	TEXT	NULLABLE	Catatan
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

4.13 Tabel pos_transaction_items

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik item
pos_transaction_id	BIGINT UNSIGNED	FK → pos_transactions(id), NOT NULL	Transaksi terkait
product_variant_id	BIGINT UNSIGNED	FK → product_variants(id), NOT NULL	Varian produk
product_name	VARCHAR(200)	NOT NULL	Snapshot nama produk
sku	VARCHAR(100)	NULLABLE	Snapshot SKU
color	VARCHAR(100)	NULLABLE	Snapshot warna
size	VARCHAR(50)	NULLABLE	Snapshot ukuran
price	DECIMAL(15,2)	NOT NULL	Harga satuan
qty	INT	NOT NULL	Jumlah
subtotal	DECIMAL(15,2)	NOT NULL	price × qty
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

FK Cascade: pos_transaction_id → ON DELETE CASCADE

4.14 Tabel email_otps

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik OTP
user_id	BIGINT UNSIGNED	FK → users(id), NULLABLE, ON DELETE SET NULL	User terkait
email	VARCHAR(255)	NOT NULL, INDEX	Email tujuan
purpose	VARCHAR(255)	NOT NULL, DEFAULT 'verify_email', INDEX	Tujuan OTP

KOLOM	TIPE	CONSTRAINT	KETERANGAN
code_hash	VARCHAR(255)	NOT NULL	Hash dari kode OTP 6 digit
attempt_count	INT UNSIGNED	NOT NULL, DEFAULT 0	Jumlah percobaan verifikasi
expires_at	TIMESTAMP	NOT NULL	Waktu kadaluarsa (+10 menit)
last_sent_at	TIMESTAMP	NULLABLE	Waktu kirim terakhir
verified_at	TIMESTAMP	NULLABLE	Waktu verifikasi sukses
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

Composite Index: (email , purpose)

4.15 Tabel promotion_media

KOLOM	TIPE	CONSTRAINT	KETERANGAN
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	ID unik media
title	VARCHAR(255)	NULLABLE	Judul media
placement	VARCHAR(255)	NULLABLE	Posisi: hero_desktop / hero_mobile / promo
active_for	VARCHAR(255)	NULLABLE	Tipe penempatan aktif
image_path	VARCHAR(255)	NOT NULL	Path file gambar
image_url	TEXT	NOT NULL	URL akses gambar
created_at	TIMESTAMP	NULLABLE	—
updated_at	TIMESTAMP	NULLABLE	—

5. Ringkasan Relasi Antar Tabel

users

- └─ (1:N) orders [user_id nullable, SET NULL]
- └─ (1:N) reviews [user_id, CASCADE]
- └─ (1:N) email_otp [user_id nullable, SET NULL]
- └─ (1:N) pos_shifts [opened_by_user_id, NO ACTION]
- └─ (1:N) personal_access_tokens [polymorphic]

categories

- └─ (1:N) products [category_id, RESTRICT delete]

products

- └─ (1:N) product_variants [product_id, CASCADE]
- └─ (1:N) product_images [product_id, CASCADE]
- └─ (1:N) order_items [product_id, CASCADE]
- └─ (1:N) reviews [product_id, CASCADE]

product_variants

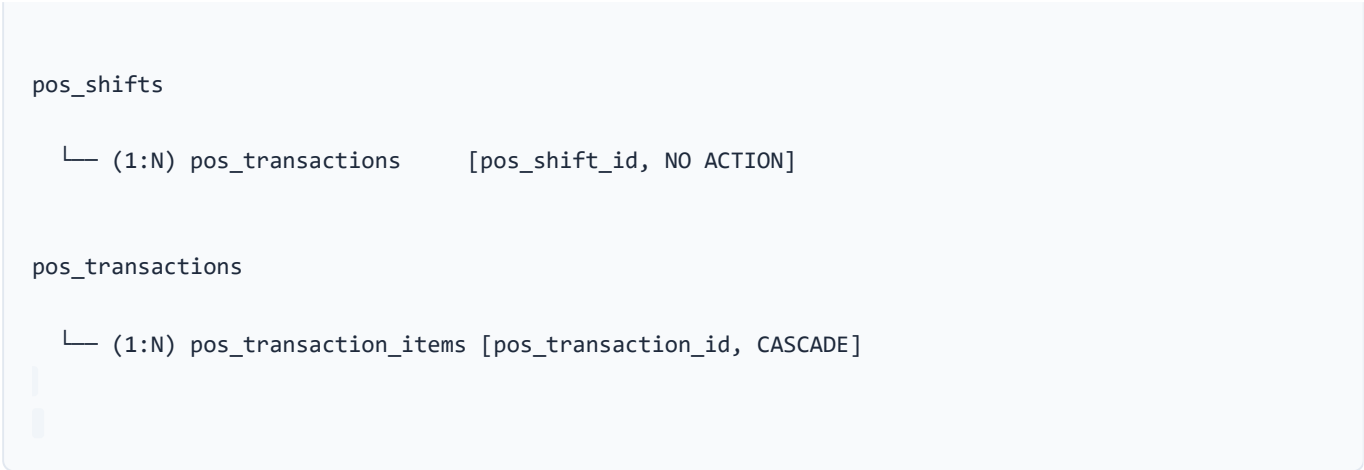
- └─ (1:N) inventories [product_variant_id, CASCADE]
- └─ (1:N) order_items [product_variant_id, CASCADE]
- └─ (1:N) pos_transaction_items [product_variant_id, NO ACTION]

orders

- └─ (1:N) order_items [order_id, CASCADE]
- └─ (1:N) reviews [order_id, CASCADE]

cashier_staff

- └─ (1:N) pos_shifts [cashier_staff_id, NO ACTION]
- └─ (1:N) pos_transactions [cashier_staff_id, NO ACTION]



6. Aturan Bisnis pada Database

#	ATURAN	IMPLEMENTASI
1	Email user harus unik dan lowercase	Unique constraint + aplikasi <code>Str::lower()</code>
2	Satu user hanya boleh satu ulasan per item order	Composite unique (<code>order_item_id</code> , <code>user_id</code>)
3	Kategori tidak bisa dihapus jika masih ada produk	ON DELETE RESTRICT pada <code>products.category_id</code>
4	Hapus produk → hapus varian, gambar, item order, ulasan	ON DELETE CASCADE
5	Hapus varian → hapus inventory	ON DELETE CASCADE
6	Hapus order → hapus semua item dan ulasan	ON DELETE CASCADE
7	User yang sudah punya order tidak bisa dihapus (data dipertahankan)	<code>orders.user_id</code> → ON DELETE SET NULL
8	Hapus transaksi POS → hapus item transaksi	ON DELETE CASCADE
9	Shift yang sedang berjalan tidak bisa dihapus	Validasi di aplikasi (PHP)
10	Stok tersedia = stok fisik - stok dipesan	Kolom <code>stock_available</code> + logika aplikasi